

V4.2 Dynamic Orbit Test v0

Based on static benchmark C3_tuned_l03_lr12_mu6

Purpose:

Test whether the calibrated static C3 field can support bounded dynamic trajectories.

Important correction:

This is not a full geodesic equation. The dynamic law below is a toy force/potential analogue built from the same radial profiles used in the static C3 model.

Dynamic potential:

$$U(r) = \text{lambda0 } R0(r) + \text{lambda_r } Rr(r) - \mu Rm(r)$$

Symbol meanings:

$U(r)$ = effective radial potential.

$R0(r)$, $Rr(r)$, $Rm(r)$ = Gaussian radial fields.

lambda0 = wide scalar delay strength = 3.

lambda_r = tight radial delay strength = 12.

μ = wide connectivity strength = 6.

Positive terms resist entry; negative term represents connectivity attraction.

Update law:

$$a = -\text{grad}(U)$$

Plain meaning:

The particle accelerates toward lower effective operational potential while keeping momentum.

Guardrail:

This is not General Relativity, not proof of gravity, and not a physical orbital derivation.

It is a dynamic graph-inspired analogue.

Best bounded case found:

$v0 = 1.000$

label = near_circular_bounded

turns = 7.94

rmin = 5.999

rmax = 6.011

radial span = 0.012

Velocity Sweep - Key Results

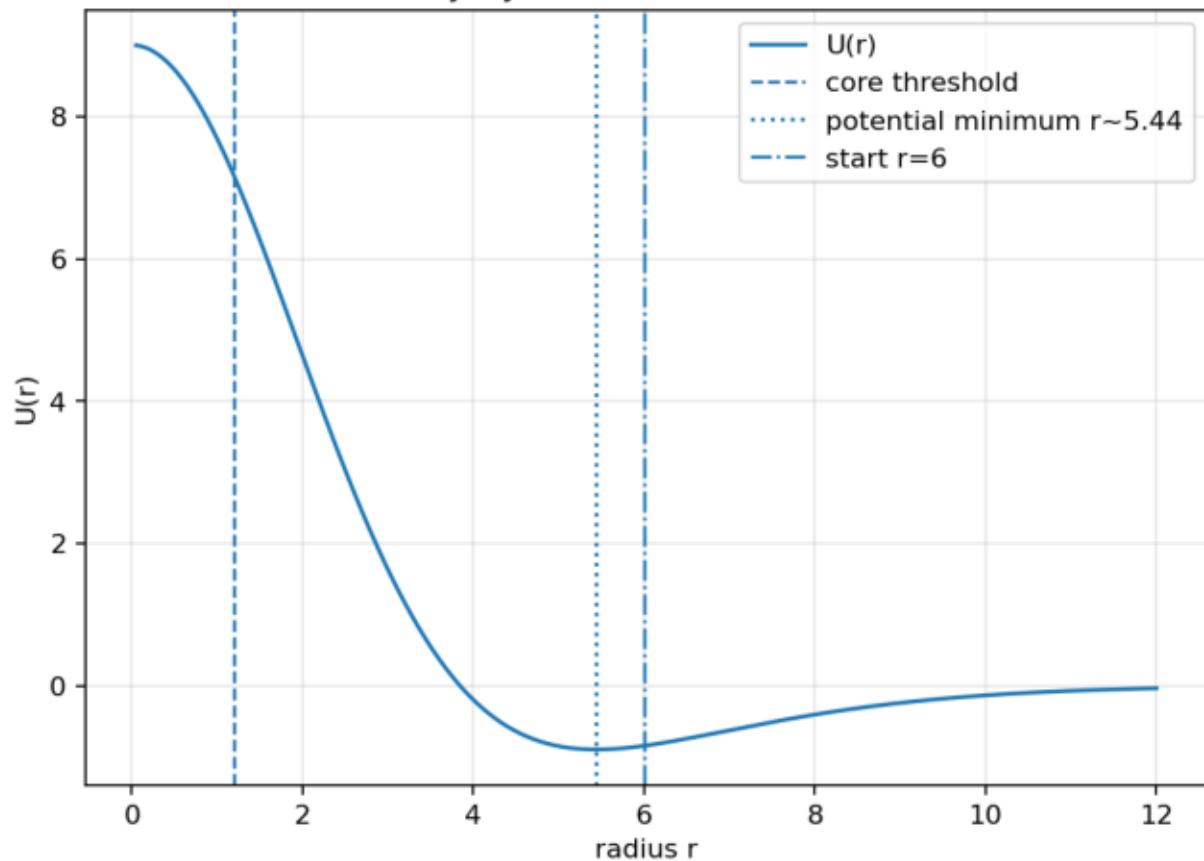
Representative cases:

v0=0.400: elliptical_bounded, r=5.13-6.00, turns=3.71, status=bounded
v0=1.000: near_circular_bounded, r=6.00-6.01, turns=7.94, status=bounded
v0=1.400: elliptical_bounded, r=6.00-7.83, turns=8.35, status=bounded
v0=1.600: escape, r=6.00-20.00, turns=0.38, status=escape

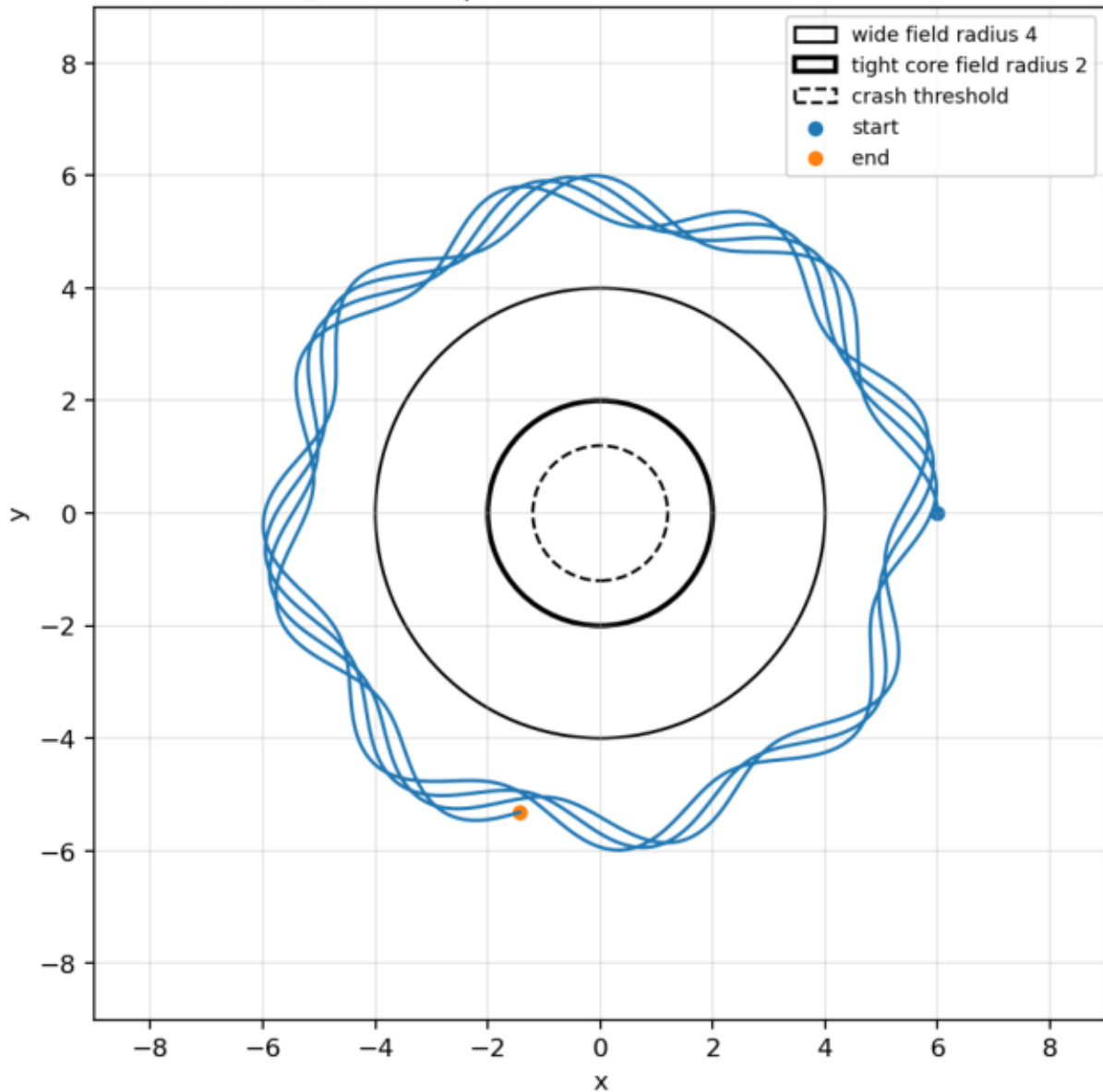
Full sweep summary:

v0=0.000, radial_oscillation	, rspan=1.024, turns=0.00, bounded
v0=0.100, radial_oscillation	, rspan=1.014, turns=0.95, bounded
v0=0.200, wide_bounded	, rspan=0.987, turns=1.90, bounded
v0=0.300, elliptical_bounded	, rspan=0.940, turns=2.82, bounded
v0=0.400, elliptical_bounded	, rspan=0.874, turns=3.71, bounded
v0=0.500, elliptical_bounded	, rspan=0.789, turns=4.57, bounded
v0=0.600, near_circular_bounded	, rspan=0.682, turns=5.37, bounded
v0=0.700, near_circular_bounded	, rspan=0.553, turns=6.13, bounded
v0=0.800, near_circular_bounded	, rspan=0.398, turns=6.81, bounded
v0=0.850, near_circular_bounded	, rspan=0.309, turns=7.13, bounded
v0=0.875, near_circular_bounded	, rspan=0.262, turns=7.27, bounded
v0=0.900, near_circular_bounded	, rspan=0.213, turns=7.42, bounded
v0=0.925, near_circular_bounded	, rspan=0.161, turns=7.56, bounded
v0=0.950, near_circular_bounded	, rspan=0.107, turns=7.69, bounded
v0=0.975, near_circular_bounded	, rspan=0.050, turns=7.82, bounded
v0=1.000, near_circular_bounded	, rspan=0.012, turns=7.94, bounded
v0=1.025, near_circular_bounded	, rspan=0.073, turns=8.06, bounded
v0=1.050, near_circular_bounded	, rspan=0.139, turns=8.16, bounded
v0=1.075, near_circular_bounded	, rspan=0.209, turns=8.26, bounded
v0=1.100, near_circular_bounded	, rspan=0.283, turns=8.35, bounded
v0=1.125, near_circular_bounded	, rspan=0.361, turns=8.43, bounded
v0=1.150, near_circular_bounded	, rspan=0.445, turns=8.51, bounded
v0=1.200, near_circular_bounded	, rspan=0.629, turns=8.62, bounded
v0=1.250, near_circular_bounded	, rspan=0.842, turns=8.68, bounded
v0=1.300, elliptical_bounded	, rspan=1.096, turns=8.67, bounded
v0=1.350, elliptical_bounded	, rspan=1.410, turns=8.58, bounded
v0=1.400, elliptical_bounded	, rspan=1.828, turns=8.35, bounded
v0=1.450, elliptical_bounded	, rspan=2.478, turns=7.80, bounded
v0=1.500, escape	, rspan=13.972, turns=0.60, escape
v0=1.550, escape	, rspan=13.950, turns=0.44, escape
v0=1.600, escape	, rspan=13.997, turns=0.38, escape
v0=1.700, escape	, rspan=14.019, turns=0.33, escape
v0=1.800, escape	, rspan=13.961, turns=0.30, escape
v0=1.900, escape	, rspan=13.939, turns=0.28, escape
v0=2.000, escape	, rspan=14.026, turns=0.27, escape

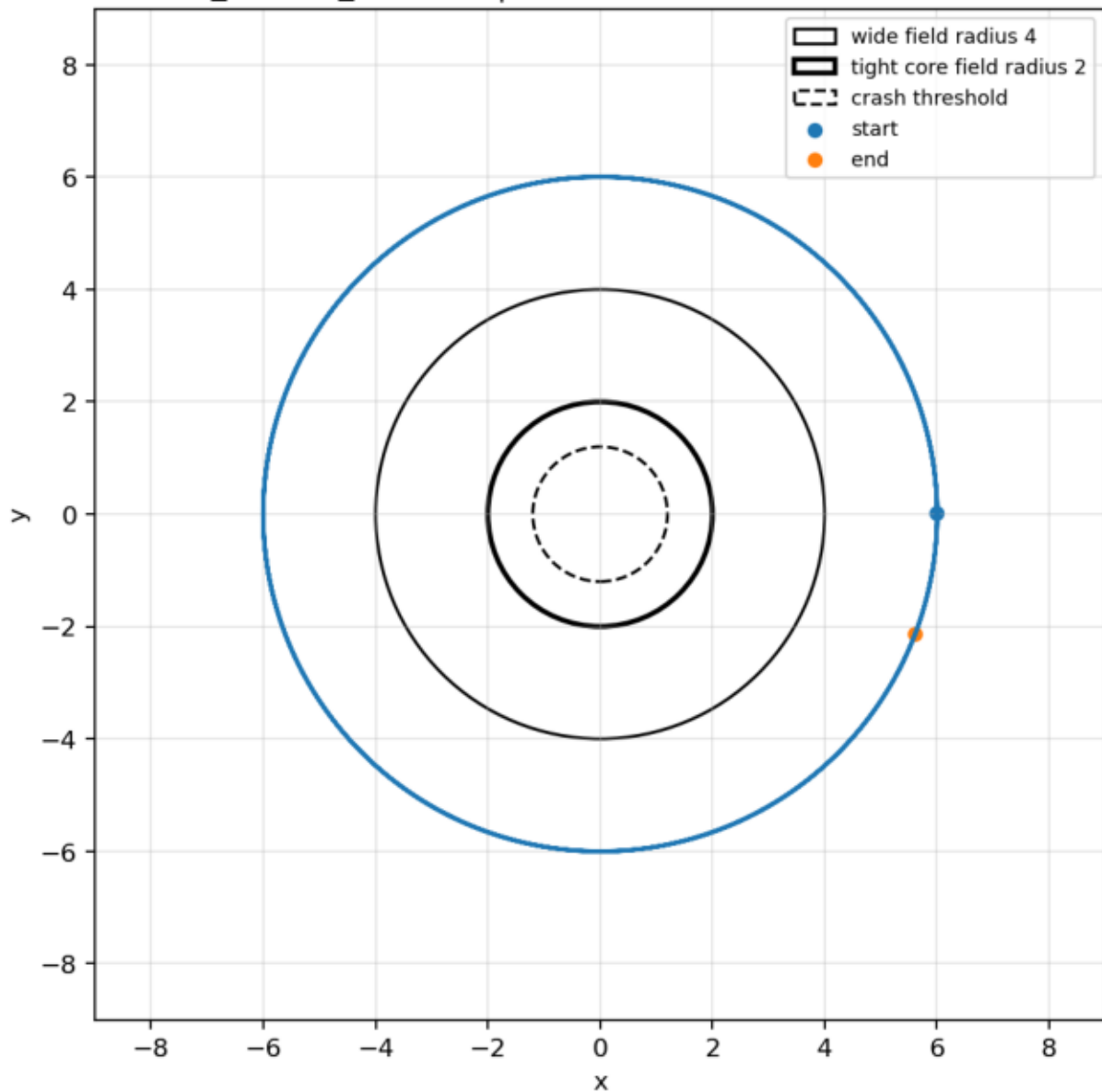
Toy Dynamic Potential Profile



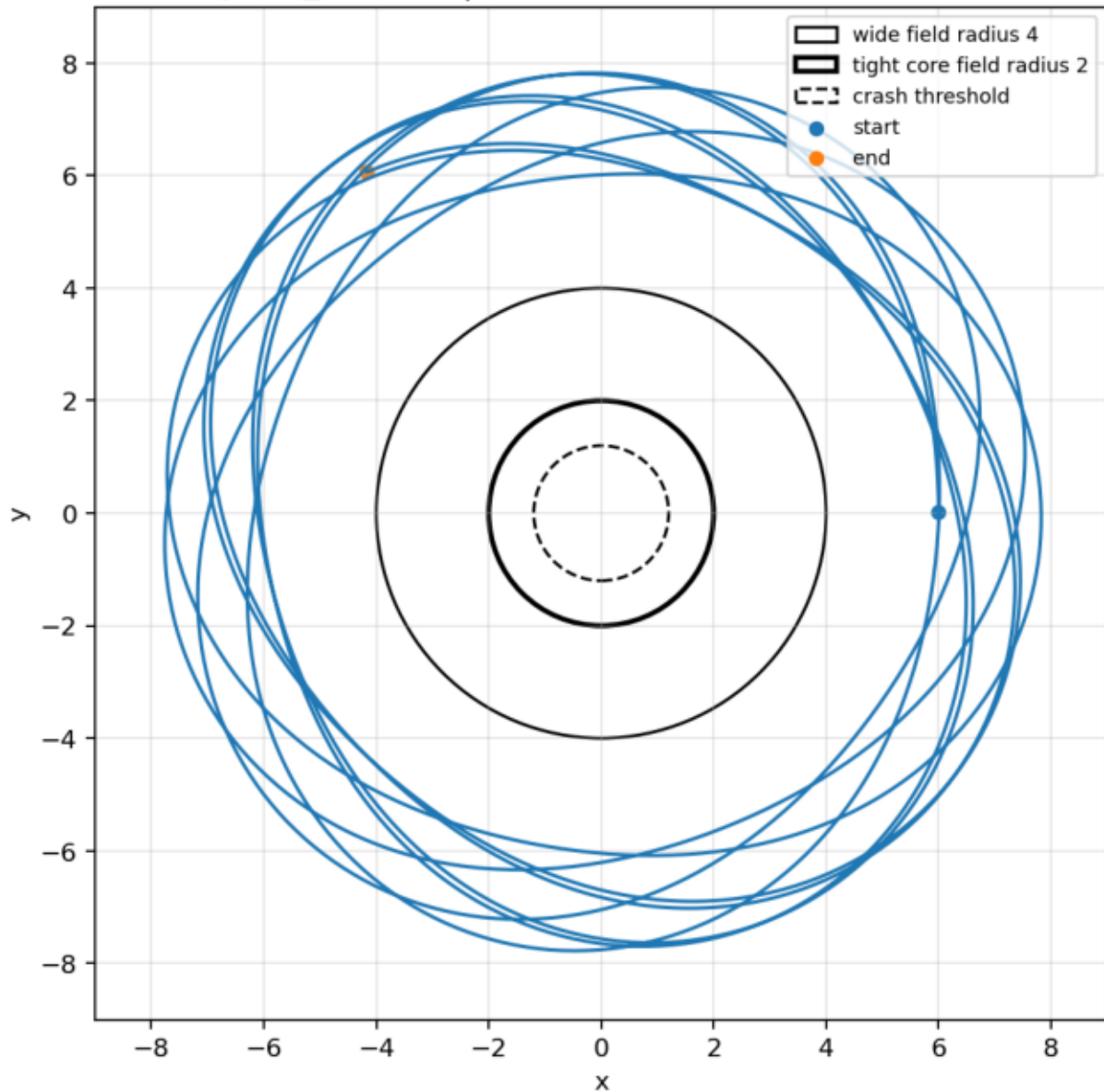
Dynamic Orbit Test v0 | $v_0=0.400$
elliptical_bounded | $r_{\min}=5.13$, $r_{\max}=6.00$, turns=3.71



Dynamic Orbit Test v0 | $v_0=1.000$
near_circular_bounded | $r_{min}=6.00$, $r_{max}=6.01$, turns=7.94



Dynamic Orbit Test v0 | $v_0=1.400$
elliptical_bounded | $r_{min}=6.00$, $r_{max}=7.83$, turns=8.35



Dynamic Orbit Test v0 | $v_0=1.600$
escape | $r_{min}=6.00$, $r_{max}=20.00$, turns=0.38

